

Dear Department of Health and Human Services Office of the Secretary,

We have prepared the following comments in response to your recent RFI: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care.

Question 1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

Response. Private-sector innovation in clinical AI is shaped less by algorithmic performance than by entrenched sociotechnical, infrastructural, and market-structure barriers. Across health systems, poor workflow fit, concentrated control of integration pathways, variable technical capacity, and ambiguity in regulatory expectations collectively slow development, prolong procurement, and reduce the viability of new entrants in the clinical AI ecosystem.

Sociotechnical and Workflow Barriers. Sociotechnical constraints consistently emerge as the most significant impediments to AI innovation and adoption. Inadequate interprofessional co-design, particularly insufficient integration of nurses, who make up the largest segment of the healthcare workforce, combined with weak alignment to real-world workflows increases cognitive burden and undermines trust, a pattern documented repeatedly across qualitative and scoping reviews.^{1,2} Health-system survey findings similarly show that administrative AI tools, such as ambient clinical documentation, achieve faster and more successful uptake than diagnostic decision-support tools because they align more naturally with existing documentation workflows and entail lower perceived clinical risk.³

Market Concentration and Vendor Dependence. Market structure further constrains innovation. Two electronic health record (EHR) vendors dominate the acute-care EHR market, jointly holding approximately two-thirds of national

¹ Hassan, M., Kushniruk, A., & Borycki, E. (2024). Barriers to and Facilitators of Artificial Intelligence Adoption in Health Care: Scoping Review. *JMIR Human Factors*, 11, e48633. <https://doi.org/10.2196/48633>

² Jacob, C., Brasier, N., Laurenzi, E., Heuss, S., Mougiakakou, S. G., Cöltekin, A., & Peter, M. K. (2025). AI for IMPACTS Framework for Evaluating the Long-Term Real-World Impacts of AI-Powered Clinician Tools: Systematic Review and Narrative Synthesis. *Journal of Medical Internet Research*, 27, e67485. <https://doi.org/10.2196/67485>

³ Poon, E. G., Lemak, C. H., Rojas, J. C., Guptill, J., & Classen, D. (2025). Adoption of artificial intelligence in healthcare: survey of health system priorities, successes, and challenges. *Journal of the American Medical Informatics Association : JAMIA*, 32(7), 1093–1100. <https://doi.org/10.1093/jamia/ocaf065>

market share, and one vendor alone accounted for nearly two-thirds of all hospital EHR purchasing decisions in 2024.^{4,5} As a result, most AI products must conform to the integration pathways, APIs, and strategic priorities of these dominant platforms, creating structural disadvantages for smaller AI developers and limiting the diversity of viable innovation pathways.⁶

Infrastructure Gaps, Technical Fragility, and Regulatory Ambiguity. Infrastructure limitations amplify these challenges in under-resourced rural and non-academic settings, where outdated IT systems, limited broadband capacity, and budget constraints hinder the computational and data-handling requirements needed for AI deployment.^{7,8} Even within well-resourced health systems, technical fragility remains a barrier: scanner- or device-related variation, batch effects, and other forms of data drift can quietly erode model performance when AI tools encounter hardware, laboratory instruments, or sequencing platforms that differ from those used during development and validation.^{7,8} Finally, regulatory ambiguity surrounding AI tools that fall outside traditional medical-device pathways contributes to prolonged procurement cycles, as legal, compliance, and risk-management teams must interpret uncertain oversight and liability frameworks. In many organizations, these reviews extend contracting timelines from several months to nearly a year, particularly for first-in-class or enterprise-scale AI deployments.^{1,3,9}

Question 2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

⁴ Landi, H. (2025). Epic gains more ground in hospital EHR market share, widens its lead over Oracle Health: KLAS. [Fierce Healthcare](#).

⁵ Pennic, F. (2025). KLAS: Epic Dominates 2024 EHR Market Share Amid Focus on Vendor Partnership; Oracle Health Sees Losses Despite Tech Advances. [HIT Consultant](#).

⁶ Marwaha, J. S., Landman, A. B., Brat, G. A., Dunn, T., & Gordon, W. J. (2022). Deploying digital health tools within large, complex health systems: key considerations for adoption and implementation. *NPJ digital medicine*, 5(1), 13. <https://doi.org/10.1038/s41746-022-00557-1>

⁷ Brown, K. E., & Davis, S. E. (2026). Gaps in artificial intelligence research for rural health in the United States: a scoping review. *Journal of the American Medical Informatics Association: JAMIA*, 33(2), 509–520. <https://doi.org/10.1093/jamia/ocaf206>

⁸ Davis, K. Telehealth and Health Information Technology in Rural Healthcare (Last Updated: 2/9/2026). [Rural Health Information Hub](#).

⁹ Mennella, C., Maniscalco, U., De Pietro, G., & Esposito, M. (2024). Ethical and regulatory challenges of AI technologies in healthcare: A narrative review. *Heliyon*, 10(4), e26297. <https://doi.org/10.1016/j.heliyon.2024.e26297>

Response. HHS should prioritize regulatory, payment, and programmatic reforms that support a total product lifecycle (TPLC) approach to clinical AI, strengthen transparency and governance, and align reimbursement with demonstrated clinical and operational value. These efforts should address gaps in post-deployment monitoring, value-based payment structures, organizational oversight, and supply-chain transparency, all of which are necessary to ensure that AI tools function safely, reliably, and equitably under real-world clinical conditions.

Regulatory Priorities. A TPLC posture is essential for both device and non-device AI tools. Consistent with emerging FDA direction for AI-enabled technologies, one-time pre-deployment accuracy assessments are insufficient; instead, regulators should require ongoing monitoring to understand how models perform and evolve in dynamic clinical environments.^{10,11} Minimum expectations should include routine performance surveillance, documentation of model versioning and provenance, including hardware and firmware dependencies, transparent change logs, and formalized drift-detection processes with documented mitigation pathways.

Payment Policy Priorities and Rewarding Verified Value. HHS should modernize payment policy to incentivize measurable improvements rather than mere technology acquisition. In alignment with the RFI's critique of fee-for-service spending bubbles, CMS should pilot value-based AI add-ons within existing payment frameworks to reimburse tools that demonstrably reduce clinician burden, expand access, or improve outcomes.¹² Such models would help shift adoption incentives away from procurement volume and toward verifiable improvements in efficiency, care quality, and patient experience.

Alignment with Federal Standards. To strengthen governance consistency across the health-care ecosystem, HHS should encourage practices analogous to the Office of Management and Budget's M-24-10 guidance, including AI inventories, named governance leadership, and minimum risk-management standards tailored for clinical use cases involving protected health information.¹³ These expectations would help health systems adopt more structured, transparent, and

¹⁰ Sandalow, M., Adams, K., & Loud, G. (2025). FDA Oversight: Understanding the Regulation of Health AI Tools. [Bipartisan Policy Center](#).

¹¹ U.S. Food and Drug Administration Total Product Life Cycle for Medical Devices. (2023). [U.S. Food and Drug Administration](#).

¹² Tierney, A. A., Reed, M. E., Grant, R. W., Doo, F. X., Payan, D. D., & Liu, V. X. (2025). Health Equity in the Era of Large Language? Models. *American Journal of Managed Care*, 31(3). <https://doi.org/10.37765/ajmc.2025.89695>

¹³ OMB M-24-10 Advancing Governance, Innovation, and Risk Management for Agency Use of Artificial Intelligence. [The White House](#).

accountable approaches to AI oversight, especially for non-device tools that fall outside traditional regulatory frameworks.

Transparency and Supply-Chain Assurance. Finally, HHS should support the creation of a standardized AI Bill of Materials (AIBOM) to improve visibility into model lineage, data sources, dependencies, and training or evaluation artifacts.¹⁴ A standardized AIBOM would build on existing federal and industry initiatives and address the growing need for transparency in AI supply chains, supporting more robust contracting, auditing, and post-market surveillance.

Question 3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

Response. Non-device AI challenges legacy accountability structures because opaque, adaptive models make it difficult to allocate liability among developers, hospitals, and clinicians, and because ‘black-box’ architectures undermine explainability and reproducibility of decision-making; in addition, when such tools influence care decisions outside clearly defined regulatory indications, they raise practical concerns about off-label use and exposure to unresolved medico-legal risk.¹⁵ Privacy questions also emerge when models infer sensitive attributes or generate secondary data that are not cleanly addressed by existing frameworks. HHS should build on emerging responsible-AI guidance by issuing model contract provisions for clinical AI, covering performance and drift-monitoring obligations, bias-mitigation duties, and notification and update service-level commitments, clarifying expectations for human oversight of non-device AI in clinical workflows, and exploring safe-harbor-style protections for organizations that implement specified governance and monitoring standards. These steps would align with broader federal AI principles emphasizing transparency, human oversight, and risk-based governance, while directly addressing the contractual and liability uncertainties that currently fuel institutional risk aversion. Clear expectations for human-in-the-loop oversight are needed to ensure that adaptive or generative AI systems remain under meaningful clinical supervision, a safeguard emphasized across recent federal AI strategy documents and FDA’s supervised agentic AI implementations.

¹⁴ Vandendriessche, W., Thijsman, J., D’hooge, L., Volckaert, B., & Sebrechts, M. (2026). AIBoMGen: Generating an AI Bill of Materials for Secure, Transparent, and Compliant Model Training. arXiv preprint arXiv:2601.05703.

¹⁵ Cestonaro, C., Delicati, A., Marcante, B., Caenazzo, L., & Tozzo, P. (2023). Defining medical liability when artificial intelligence is applied on diagnostic algorithms: a systematic review. *Frontiers in Medicine*, 10, 1305756. <https://doi.org/10.3389/fmed.2023.1305756>

Question 4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?

Response. Robust evaluation must extend beyond accuracy to encompass workflow fit, human-AI teaming, and safety. Pre-deployment, health systems should use shadow-mode comparisons, usability and workload testing, and bias assessment across subgroups to detect risks before tools influence care. Evaluation frameworks should also assess the integrity of human-in-the-loop workflows, including how clinicians verify, override, and contextualize AI-generated recommendations, and how these oversight points affect safety, workflow reliability, and cognitive load under real-world conditions. These human-factors assessments should explicitly evaluate cognitive load impacts and identify temporal clustering effects (e.g., whether shift-change patterns or simultaneous user activity degrade model responsiveness or reliability).² Post-deployment, organizations need formal drift-monitoring frameworks that continuously track input data-distribution shifts, model performance, and key human-AI interaction signals, such as clinician overrides, building on approaches already piloted for radiology and other imaging AI tools.^{9,16} Federal advisory discussions similarly emphasize evaluating generative AI for sociotechnical impact (workflow disruption, accountability, safety), not accuracy alone.¹⁷ HHS can accelerate rigor by funding multi-site evaluation testbeds, issuing human-factors guidance, and deploying cooperative agreements or prize competitions to mature standardized benchmarking.

Question 5. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

Response. Successful adoption correlates with governance maturity, leadership alignment, and cross-stakeholder collaboration, which are dimensions that can be operationalized through standardized organizational readiness and maturity models for

¹⁶ Cook, C. J., Klug, J. R., Kandler, B. W., Baez-Suarez, A., Dachowicz, A. P., Blezek, D. J., . . . Kline, T. L. (2026). State of the AI: Post-Deployment Monitoring of Radiology-Focused Internally Developed AI. *Mayo Clinic Proceedings: Digital Health*, 100342.

<https://doi.org/https://doi.org/10.1016/j.mcpdig.2026.100342>

¹⁷ Cestonaro, C., Delicati, A., Marcante, B., Caenazzo, L., & Tozzo, P. (2023). Defining medical liability when artificial intelligence is applied on diagnostic algorithms: a systematic review. *Frontiers in Medicine*, 10, 1305756. <https://doi.org/10.3389/fmed.2023.1305756>

clinical AI.¹⁸ HHS should recognize, or help convene, third-party certification and independent testing programs that evaluate safety, bias, robustness, including hardware/firmware variability, and workflow outcomes using common reporting artifacts (e.g., model cards plus AIBOM-style disclosures). Accreditation and certification efforts should be complemented by workforce-level competency development to ensure that organizational governance expectations are matched by staff skills and readiness. Targeted workforce micro-credentialing for clinicians, informaticians, and operational leaders would further professionalize AI governance and reduce adoption risk.

Question 7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

Response. Artificial intelligence adoption in clinical care is shaped not only by technological performance but also by governance structures, workforce readiness, and implementation strategy. Importantly, responsible AI use does not end at initial implementation and must be integrated into the daily, ongoing operational work of healthcare organizations.

Key Point 1-AI Adoption Requires Structured Governance and Oversight

Adoption of AI tools within healthcare organizations depends on clearly defined governance pathways that involve executive leadership, clinical leadership, informatics teams, compliance officers, and frontline stakeholders. The National Academy of Medicine emphasizes that safe and effective integration of AI requires institutional oversight mechanisms, defined accountability structures, and transparency in system design and deployment.¹⁹ Similarly, the World Health Organization underscores the importance of regulatory clarity, lifecycle monitoring, and institutional governance in its guidance on artificial intelligence for health.²⁰ Without structured oversight, organizations may hesitate to adopt AI due to uncertainty regarding validation standards, risk management, and responsibility. As a best practice, federal guidance could encourage patient-inclusive governance structures, clear accountability for AI oversight

¹⁸ Nair, M., Svedberg, P., Larsson, I., & Nygren, J. M. (2024). A comprehensive overview of barriers and strategies for AI implementation in healthcare: Mixed-method design. *PloS One*, 19(8), e0305949. <https://doi.org/10.1371/journal.pone.0305949>

¹⁹ National Academy of Medicine. (2023). *Artificial intelligence in health care: The hope, the hype, the promise, the peril*. National Academies Press (US). <https://doi.org/10.17226/27111>

²⁰ World Health Organization. (2023). *Regulatory considerations on artificial intelligence for health*. <https://www.who.int/publications/i/item/9789240078879>

within leadership roles, and organizational strategies that explicitly address transparency, safety, and ethical use of AI across clinical and administrative functions.

Key Point 2- Workforce Readiness and Competency Development are Critical

A significant administrative hurdle to AI adoption is the preparedness of healthcare professionals to evaluate, interpret, and appropriately integrate AI-supported outputs into clinical reasoning and documentation. The National Academy of Medicine highlights workforce training and competency development as essential components of responsible AI integration.²⁰ Workforce readiness directly shapes documentation transparency, communication with patients, and decision-making confidence. Without structured education and clear expectations for clinician oversight, organizations may experience hesitancy, inconsistent use, or inappropriate reliance on AI-supported tools. Investment in competency frameworks and practical training is therefore central to accelerating safe and sustainable adoption. This need for workforce preparation is ongoing and “evergreen,” particularly given routine staff turnover, the growth of contingent workforce models, and the critical role of nonclinical staff who frequently serve as primary interfaces with patients in both hospital and outpatient settings.

Key Point 3 - Implementation Challenges Extend Beyond Technical Performance

AI adoption barriers frequently arise not from algorithm performance alone but from workflow disruption, ambiguity regarding liability, change-management demands, and integration into existing documentation and quality oversight processes. Implementation efforts must address how AI tools fit within clinical workflows, how responsibility is allocated when AI informs decisions, and how AI involvement is reflected in documentation without compromising clarity or compliance. Regulatory developments such as the ONC’s Health Data, Technology, and Interoperability (HTI) rulemaking, including HTI-1 provisions on algorithm-transparency requirements for certified health IT, highlight the growing importance of documentation disclosure and explainability in operational settings.²¹ Addressing these nontechnical and operational factors is essential to translating AI innovation into sustainable clinical practice.

Clear federal guidance that supports multidisciplinary governance, workforce competency development, and transparent documentation standards may help reduce institutional uncertainty and accelerate the responsible adoption of AI in clinical care.

²¹ Office of the National Coordinator for Health Information Technology. (2023). *Health data, technology, and interoperability: Certification program updates, algorithm transparency (HTI-1 Final Rule)*.

Question 9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

Response. Patients and caregivers generally appreciate the potential of artificial intelligence in clinical care to improve understanding, access, and coordination of information, but meaningful adoption hinges on addressing concerns regarding trust, transparency, explainability, and accountability.

Key Point 1- Improve Understanding and Navigation

Patients and caregivers frequently encounter challenges in interpreting clinical documentation, coordinating information across care teams, and obtaining clear explanations of their conditions and treatments. AI-enabled tools that summarize clinical content, enhance clarity, and support navigation through health records hold promise for addressing these challenges. Research indicates that a majority of surveyed hospital patients, across multiple countries, expressed generally positive overall attitudes toward AI use in health care, suggesting receptivity to technologies that improve comprehension and accessibility when they are integrated thoughtfully into care processes.²²

Key Point 2- Concerns Related to Trust, Accountability, and Equity

Despite general openness to AI's potential, patients express caution about specific aspects of AI use. The study by Busch and colleagues (2025) found that patients preferred AI systems that are explainable and that clinical decision-making remains under physician oversight rather than autonomous AI control. Attitudes varied by demographic, health status, and technological literacy, with notable skepticism about AI's accuracy in some diagnostic information. These findings highlight that trust in AI is variable and is strongly influenced by transparency, clinician involvement, and appropriate tailoring of AI implementation.

Key Point 3- Transparency and Interpretability Must Guide Deployment

As patients increasingly interact with electronic health records and clinical information, transparency and interpretability become essential for engagement and trust. Evidence shows that patients are not uniformly confident in all aspects of AI use, underscoring the need for clear communication about how AI is applied, what it contributes, and how human clinicians remain responsible for

²² Busch, F., Hoffmann, L., Xu, L., et al. (2025). Multinational attitudes toward AI in health care and diagnostics among hospital patients. *JAMA Network Open*, 8(6), e2514452. <https://doi.org/10.1001/jamanetworkopen.2025.14452>

oversight. Prioritizing systems that clearly indicate AI involvement, explain reasoning, and reinforce clinician validation may enhance patient comfort and support equitable participation in care. Aligning implementation strategies with patient preferences for explainability and oversight can help ensure that AI adoption strengthens rather than undermines patient experience. Patients and caregivers may expect AI-supported care to align with principles of shared decision-making, particularly when AI-informed documentation and clinical recommendations are visible through patient-facing records and portals.

Policies and implementation strategies that prioritize transparency, clinician oversight, and clear communication about AI involvement may help ensure that adoption strengthens patient trust and engagement rather than introducing uncertainty or confusion.

Question 10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care? a. Are there published findings about the impact of adopted AI tools and their use in clinical care? b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?

Response. HHS should prioritize research that strengthens the real-world performance, safety, and interoperability of clinical AI systems across diverse care environments. Evidence from U.S. health-system surveys shows that AI tools succeed when they reduce documentation burden and integrate smoothly into workflows, whereas higher-risk diagnostic tools face more limited uptake, suggesting that research on implementation science, human-AI teaming, and workflow fit remains foundational for improving adoption. A second priority concerns rigorous evaluation frameworks that capture long-term clinical, organizational, and economic impact. The AI for IMPACTS systematic review highlights the need for multidimensional evaluation, encompassing integration, governance, safety, cost, and scalability, which expands beyond accuracy metrics and aligns closely with a total-product-lifecycle approach increasingly emphasized in federal policy.^{2,23} Third, as large language models and emerging multi-modal AI systems become embedded in documentation, summarization, and decision-support tasks, research should assess their reliability, error modes, human-oversight requirements, and performance across subpopulations and care settings. Research is also needed to define optimal human-in-the-loop models for large language models, agentic systems, and multimodal AI, determining when human review is required, how oversight affects safety and equity, and how to balance automation

²³ Kiyasseh, D., Cohen, A., Jiang, C., & Altieri, N. (2024). A framework for evaluating clinical artificial intelligence systems without ground-truth annotations. *Nature Communications*, 15(1), 1808. <https://doi.org/10.1038/s41467-024-46000-9>

benefits with clinician accountability. Fourth, HHS should support studies addressing robustness and context transfer, as peer-reviewed findings show that variations in scanner vendors, acquisition protocols, and biomedical data platforms can meaningfully degrade model performance, highlighting the need for harmonization strategies and drift-monitoring frameworks to ensure equitable deployment across resource-variable sites. Finally, HHS should invest in economic and operational evaluations that quantify the value of AI in clinical practice. Despite widespread deployment, few studies systematically measure time returned to patient care, downstream utilization effects, or total cost of ownership. Early evidence suggests that administrative AI offers clearer near-term benefits than diagnostic tools, but rigorous multi-site assessment is required to inform reimbursement models and ensure that financial incentives align with verifiable improvements in efficiency, quality, and access.

10(a). Published findings about the impact of adopted AI tools

Published data from national health-system surveys show that ambient documentation and other workflow-support tools generate the most consistent early improvements, reducing documentation time, lowering burden, and improving satisfaction, while diagnostic AI tools demonstrate more variable performance and lower perceived success.^{3,24} Review evidence further indicates that long-term impact depends on sustained monitoring of governance, usability, safety, and economic outcomes rather than one-time accuracy evaluations.²⁵

10(b). Costs, benefits, and transfers in the literature

The literature approaches the economics of AI cautiously: a recent systematic review of clinical AI economic evaluations found only a small number of formal cost-effectiveness and budget-impact studies compared with the volume of clinical AI trials, with substantial heterogeneity in methods and frequent omission of key cost components. Most published analyses emphasize near-term operational efficiencies, such as reductions in unnecessary procedures or screening costs, rather than comprehensive assessments that fully account for acquisition, integration, maintenance, downstream outcomes, and distributional effects. Existing reviews and actuarial analyses therefore call for standardized approaches to quantify operational savings, avoided adverse events, and

²⁴ Lukac, P. J., Turner, W., Vangala, S., Chin, A. T., Khalili, J., Shih, Y. T., Sarkisian, C., Cheng, E. M., & Mafi, J. N. (2025). Ambient AI Scribes in Clinical Practice: A Randomized Trial. *NEJM AI*, 2(12), 10.1056/aioa2501000. <https://doi.org/10.1056/aioa2501000>

²⁵ Nguyen, T. D., Whaley, C. M., Simon, K., Mehta, N., Yu, H., McBain, R. K., Mehrotra, A., & Cantor, J. H. (2025). Adoption of Artificial Intelligence in the Health Care Sector. *JAMA Health Forum*, 6(11), e255029. <https://doi.org/10.1001/jamahealthforum.2025.5029>

workforce redistribution, and to clarify how AI-generated value is allocated among clinicians, health systems, payers, and patients.²⁶

²⁶ El Arab, R. A., & Al Moosa, O. A. (2025). Systematic review of cost effectiveness and budget impact of artificial intelligence in healthcare. *NPJ Digital Medicine*, 8(1), 548.
<https://doi.org/10.1038/s41746-025-01722-y>

Sincerely,

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